

Reset Password in ASP.NET Core

To reset password

- Email
 - Password reset token
 - New Password &
 - Confirm Password
- } In the password reset link
- } User provides

Reset Password - View & View Model

4 properties:

- Email
- Password &
- Confirm Password
- Password reset token

```
public class ResetPasswordViewModel
{
    [Required]
    [EmailAddress]
    public string Email { get; set; }

    [Required]
    [DataType(DataType.Password)]
    public string Password { get; set; }

    [DataType(DataType.Password)]
    [Display(Name = "Confirm password")]
    [Compare("Password", ErrorMessage = "Password and Confirm Password must match")]
    public string ConfirmPassword { get; set; }

    public string Token { get; set; }
}
```

```
using System.ComponentModel.DataAnnotations;
using System.Xml.Linq;
```

```
namespace EmployeeManagement.ViewModels
{
    public class ResetPasswordViewModel
    {
        [Required]
        [EmailAddress]
        public string Email { get; set; }

        [Required]
        [DataType(DataType.Password)]
        public string Password { get; set; }

        [DataType(DataType.Password)]
        [Display(Name = "Confirm password")]
        [Compare("Password",
            ErrorMessage = "Password and Confirm Password must match")]
        public string ConfirmPassword { get; set; }
    }
}
```

```

    public string Token { get; set; }
}
}

```

Reset Password View & View Code

Reset Password

```

<form method="post">
  <div asp-validation-summary="All" class="text-danger"></div>
  <input asp-for="Token" type="hidden" />
  <input asp-for="Email" type="hidden" />
  <div class="form-group">
    <label asp-for="Password"></label>
    <input asp-for="Password" class="form-control" />
    <span asp-validation-for="Password" class="text-danger"></span>
  </div>
  <div class="form-group">
    <label asp-for="ConfirmPassword"></label>
    <input asp-for="ConfirmPassword" class="form-control" />
    <span asp-validation-for="ConfirmPassword" class="text-danger"></span>
  </div>
  <button type="submit" class="btn btn-primary">Reset</button>
</form>

```

View→Account→ResetPassword.cshtml

@model ResetPasswordViewModel

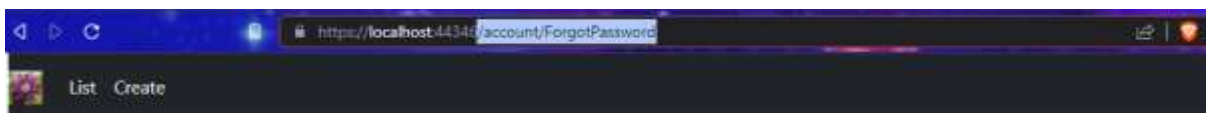
```

<h2>Reset Password</h2>
<hr />
<div class="row">
  <div class="col-md-12">
    <form method="post">
      <div asp-validation-summary="All" class="text-danger"></div>
      <input asp-for="Token" type="hidden" />
      <input asp-for="Email" type="hidden" />
      <div class="form-group">
        <label asp-for="Password"></label>
        <input asp-for="Password" class="form-control" />
        <span asp-validation-for="Password" class="text-danger"></span>
      </div>
      <div class="form-group">
        <label asp-for="ConfirmPassword"></label>
        <input asp-for="ConfirmPassword" class="form-control" />
        <span asp-validation-for="ConfirmPassword" class="text-danger"></span>
      </div>
      <button type="submit" class="btn btn-primary">Reset</button>
    </form>
  </div>
</div>

```

AccountController:

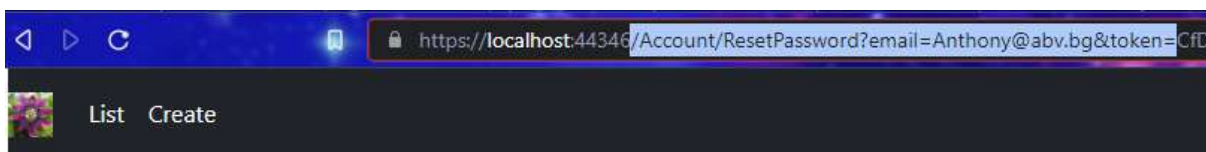
```
[HttpGet]
[AllowAnonymous]
public IActionResult ResetPassword(string token, string email)
{
    // If password reset token or email is null, most likely the
    // user tried to tamper the password reset link
    if (token == null || email == null)
    {
        ModelState.AddModelError("", "Invalid password reset token");
    }
    return View();
}
```



If you have an account with us, we have sent an email with the instructions to reset your password.

C:\DemoLogs

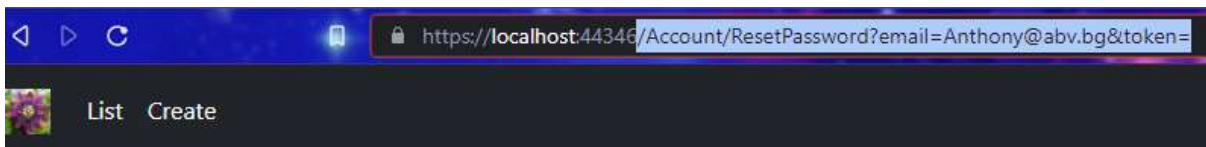
<https://localhost:44346/Account/ResetPassword?email=Anthony@abv.bg&token=CfDJ8NXl8GzyYcNDu0rWycevlizOli%2FTx4Zy%2BjSE6qOfgkOihqquoM61F8V1VdL6mE89mof8Gx4xUfWjwAYi9Pbiq3QIM0DFsC2Wo3sfxxOHMnZltkaHM3IfTXF9ocdKxQaMM7OlzaleRWYj6BP2D9kNf8NCUcUD2c%2BFQ%2B29N1cEEulslvuLMo6ZRVE1Tlpeq87gR9nWljR%2F4UPKTIAe4blJyAZ4XMCKbRb5n49jdYPgWRx>



Reset Password

Form fields for Password and Confirm password, with a Reset button.

Ако изтрием токена или имейла, if-а няма да бъде изпълнен:



Reset Password

• Invalid password reset token

Ако ги възстановим, валидацията изчезва.

Ако въведем парола и дадем Reset, връща 404 за post на ResetPassword() action.

Reset Password Action

```
public async Task<IActionResult> ResetPassword(ResetPasswordViewModel model)
{
    if (ModelState.IsValid) password, confirmPassword, email, token
    {
        var user = await userManager.FindByEmailAsync(model.Email);
        if (user != null) if the user is found
        {
            var result = await userManager.ResetPasswordAsync(user, model.Token, model.Password);
            if (result.Succeeded)
            {
                return View("ResetPasswordConfirmation");
            }
            foreach (var error in result.Errors)
            {
                ModelState.AddModelError("", error.Description);
            }
            return View(model); with all errors
        }
    }
    else return View("ResetPasswordConfirmation"); without message against bruteforce attacks
    return View(model);
}
```

Интересното тук е, че ако не се намери потребител с дадения имейл, не се извежда съобщение, умишлено, за да се намалят шансовете за brute-force атака, а просто се връща страницата за ResetPasswordConfirmation.

[HttpPost]

[AllowAnonymous]

```
public async Task<IActionResult> ResetPassword(ResetPasswordViewModel model)
{
```

```
    if (ModelState.IsValid)
    {
        // Find the user by email
        var user = await userManager.FindByEmailAsync(model.Email);

        if (user != null)
        {
            // reset the user password
            var result = await userManager.ResetPasswordAsync(user, model.Token,
model.Password);
            if (result.Succeeded)
            {
                return View("ResetPasswordConfirmation");
            }
            // Display validation errors. For example, password reset token already
            // used to change the password or password complexity rules not met
            foreach (var error in result.Errors)
            {
                ModelState.AddModelError("", error.Description);
            }
            return View(model);
        }
    }
}
```

```
// To avoid account enumeration and brute force attacks, don't
// reveal that the user does not exist
```

```

        return View("ResetPasswordConfirmation");
    }
    // Display validation errors if model state is not valid
    return View(model);
}

```

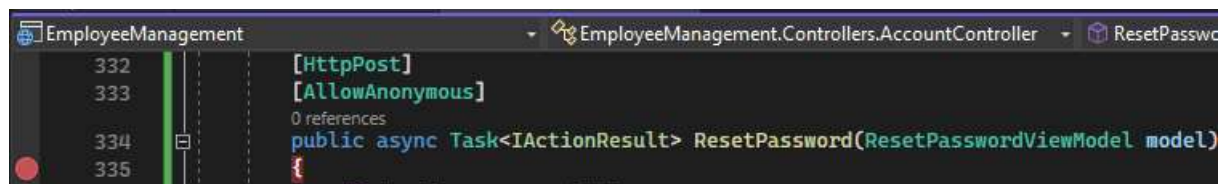
ResetPasswordConfirmation.cshtml

```

<h4>
    Your password is reset. Please click <a asp-action="Login">here to login</a>
</h4>

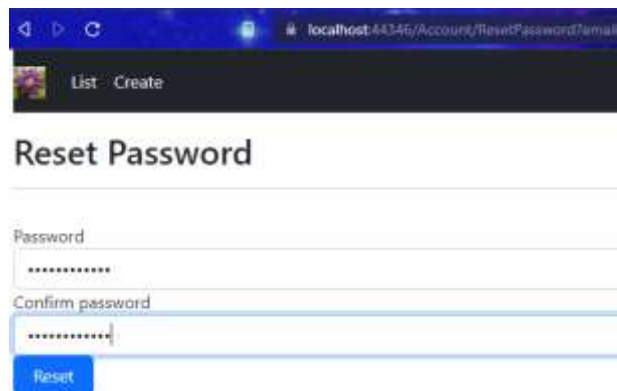
```

Експериментално включваме Debug mode ->

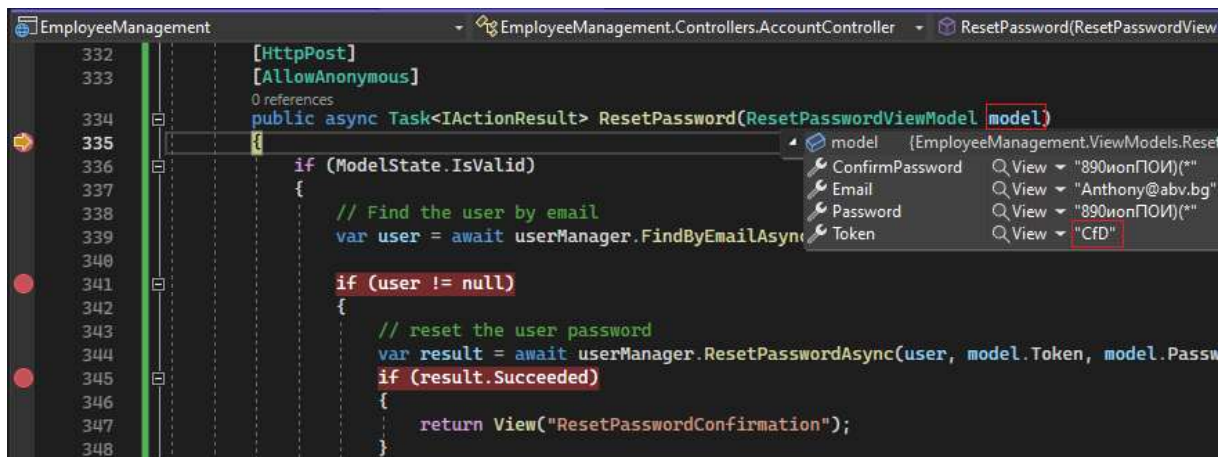


<https://localhost:44346/Account/ResetPassword?email=Anthony@abv.bg&token=CfDJ8NXl8GzyYcNDu0rWyceVLizOli%2FTrx4Zy%2BjSE6qOfgkOihqquoM61F8V1VdL6mE89mof8Gx4xUfWjwAYi9Pbiq3QIM0DFsC2Wo3sfxOHMnZltkaHM3IfTXF9ocdKxQaMM70IzaleRWYj6BP2D9kNf8NCUcUD2c%2BFQ%2B29N1cEEulsIvuLMo6ZRVE1Tlpeq87gR9nWijR%2F4UPKTIAe4bJyAZ4XMCKbRb5n49jdYPgwRx>

премахваме последните символи (за по-добра видимост, аз оставих само 3 символа от токена token=CfD), изгледът остава същия след refresh и въвеждаме нова парола, отговаряща на всички условия за сложност:



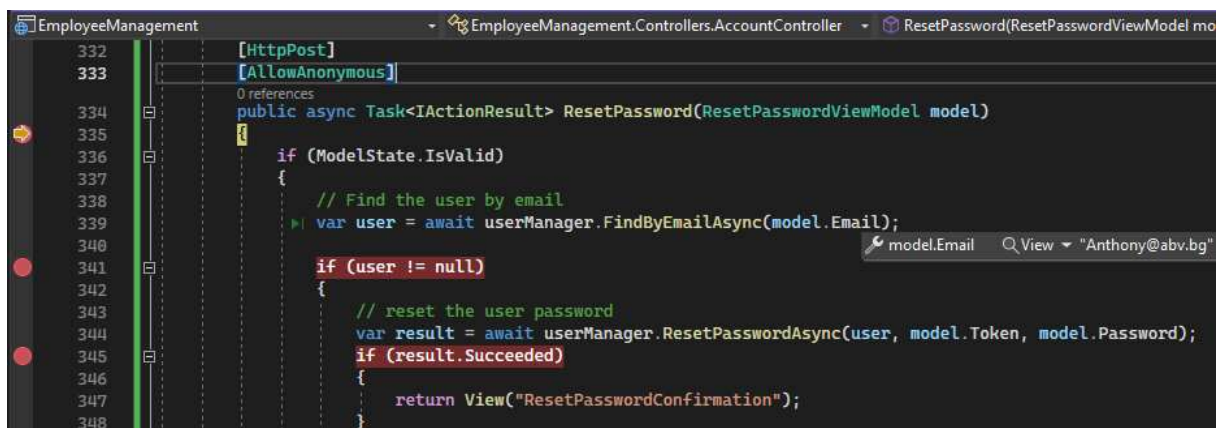
BP1 – извлечени данни, включително скъсения токен:



```
[HttpPost]
[AllowAnonymous]
public async Task<IActionResult> ResetPassword(ResetPasswordViewModel model)
{
    if (ModelState.IsValid)
    {
        // Find the user by email
        var user = await userManager.FindByEmailAsync(model.Email);

        if (user != null)
        {
            // reset the user password
            var result = await userManager.ResetPasswordAsync(user, model.Token, model.Password);
            if (result.Succeeded)
            {
                return View("ResetPasswordConfirmation");
            }
        }
    }
}
```

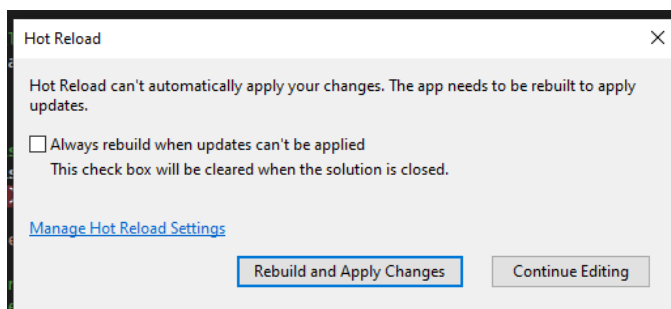
Имейлът е разпознат -> BP2



```
[HttpPost]
[AllowAnonymous]
public async Task<IActionResult> ResetPassword(ResetPasswordViewModel model)
{
    if (ModelState.IsValid)
    {
        // Find the user by email
        var user = await userManager.FindByEmailAsync(model.Email);

        if (user != null)
        {
            // reset the user password
            var result = await userManager.ResetPasswordAsync(user, model.Token, model.Password);
            if (result.Succeeded)
            {
                return View("ResetPasswordConfirmation");
            }
        }
    }
}
```

При мен обаче, user остава неразпознат и извежда грешка:



При старт без Debug ->

Следваща проверка в Debug mode е с истинския токен и парола, неотговаряща на правилата „123“:

BP1:

```

334 public async Task<IActionResult> ResetPassword(ResetPasswordViewModel model)
335 {
336     ConfirmPassword = "123"
337     Email = "Anthony@abv.bg"
338     Password = "123"
339     Token = "CfDJ8NXI8GzyYcNDu0rWyceVLizOli/Trx4Zy+jSE6qOfgkOihqquoM61F8V1VdL6mE89mof8C"

```

BP2:

```

341 if (user != null)
342 {

```

BP3:

```

341 if (user != null)
342 {
343     // reset the user password
344     var result = await userManager.ResetPasswordAsync(user, model.Token, model.Password);
345     if (result.Succeeded)
346     {
347         return View("ResetPasswordConfirmation");

```

Имаме множество валидационни грешки, които се виждат в .Succeeded.

```

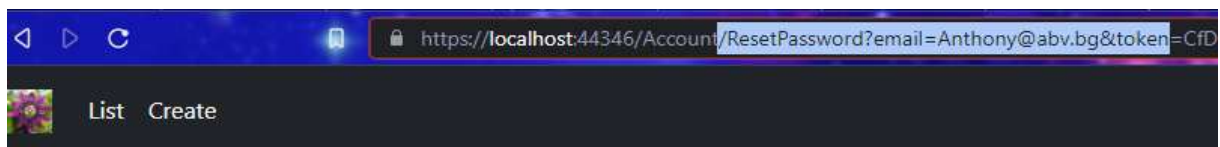
345 if (result.Succeeded)
346 {

```

Continue:

Въвеждаме нова валидна парола и отново ни изхвърля към BP1 ->

```
334 public async Task<IActionResult> ResetPassword(ResetPasswordViewModel model)
335 {
336     ConfirmPassword Q View "123qweQWE!@#"
337     Email Q View "Anthony@abv.bg"
338     Password Q View "123qweQWE!@#"
339     Token Q View "CfDJ8NXI8GzyYcNDu0rWycevLizOli/Txx4Zy+jSE6qOfgkOihqquoM61F8V1VdL6mE89mof8Gx4xUfWjwAYt9Pbiq3QIM0DFs"
340
341     if (user != null)
342     {
343         // reset the user password
344         var result = await userManager.ResetPasswordAsync(user, model.Token, model.Password);
345         if (result.Succeeded) ≤ 523ms elapsed
346         {
347             return View("ResetPasswordConfirmation");
348         }
349     }
350 }
```



Your password is reset. Please click [here to login](#)

Въвеждаме новата парола и се вписваме (за по-голяма сигурност, това не става автоматично).